Algorithm 1: Binary Atom Search Optimization (BASO) with True PRLU

Input: Population size N, Max iterations T, Parameters α, β, Dataset with F features.

Output: Optimal binary feature subset $\mathbf{X}_{gb}$.

Initialize binary population $\mathbf{X} \in \{0,1\}^{N \times F}$ randomly;
Initialize velocities $\mathbf{V} = \mathbf{0}^{N \times F}$ and set $V_{max} = 6$;
Evaluate fitness for all atoms and find global best $\mathbf{X}_{gb}$ and Fit_{gb};
for $t = 1$ **to** T **do**
    Calculate mass $M_i(t)$ for all atoms using Eq. (2);
    Determine $K_{best}(t)$ using Eq. (3);
    Update time-varying parameters $\eta(t)$ and $\lambda(t)$ using Eq. (4) and (5);
    for *each atom* $i = 1$ **to** N **do**
        Calculate length scale σ_i relative to centroid of K_{best} atoms (Eq. 6);
        $\mathbf{A}_i \leftarrow \mathbf{0}$; `// Initialize acceleration`
        for *each atom* $j \in K_{best}$ **do**
            Calculate Euclidean distance $r_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\|$;
            Calculate normalized distance $h_{ij} = \text{clamp}\left(\frac{r_{ij}}{\sigma_i + \epsilon}, h_{min}, h_{max}\right)$;
            Compute Lennard-Jones force magnitude $F_{mag} = -\eta(t) \cdot (2h_{ij}^{13} - h_{ij}^{7})$;
            Generate random weight $r_j \sim U(0,1)$;
            $\mathbf{A}_i \leftarrow \mathbf{A}_i + r_j \cdot F_{mag} \frac{\mathbf{X}_j - \mathbf{X}_i}{r_{ij} + \epsilon}$;
        `// Add global best attraction and divide by mass (Eq. 9)`
        $\mathbf{A}_i \leftarrow \frac{\mathbf{A}_i + \lambda(t)(\mathbf{X}_{gb} - \mathbf{X}_i)}{M_i(t) + \epsilon}$;
        for *each dimension* $d = 1$ **to** F **do**
            `// Update velocity (Eq. 10)`
            $V_i^d \leftarrow \text{clip}\left(rand(0,1) \cdot V_i^d + A_i^d, -V_{max}, V_{max}\right)$;
            `// Map to binary position using True PRLU`
            Calculate probability $S = f_{\text{TruePRLU}}(V_i^d)$;
            $X_i^d \leftarrow 1$ **if** $rand(0,1) < S$ **else** 0;
    `// Re-evaluate fitness and update global best`
    Evaluate fitness for all atoms;
    if *new best fitness found* **then**
        Update $\mathbf{X}_{gb}$ and Fit_{gb};
return $\mathbf{X}_{gb}$;